

Learning Urban Microclimate Dynamics via Reduced Order Modeling from Multiphysics Simulations

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Abstract—This study presents a Reduced Order Modeling (ROM) framework for predicting high-resolution urban microclimate fields based on outputs from multiphysics simulations. Using Random Forest and Gradient Boosting regressors trained on CFD-generated data, the model accurately predicts spatial distributions of air temperature, wind velocity magnitude, and mean radiant temperature (MRT) across diverse weather conditions. Dimensionality reduction via PCA and targeted input rebalancing were applied to enhance generalization and performance. The resulting ROM reduces inference time from 25 minutes per CFD timestep to just 7 seconds, enabling rapid scenario evaluation. Despite strong overall accuracy, the model assigned low importance to wind direction, indicating a limitation in capturing directional flow dynamics—particularly when extrapolating to unknown wind approach angles. The proposed approach offers an interpretable and computationally efficient tool for urban climate analysis relevant to human thermal experience.

CODE REPOSITORY

https://github.com/matzeFIT/ROM-UPC_BCN.git

Index Terms—Reduced Order Model, Random Forest, Urban Microclimate, Thermal Comfort, CFD, Physics-Informed Machine Learning.

I. INTRODUCTION

Prototyping the local atmosphere is essential for advancing the design of human-centered living spaces, especially in the context of increasing weather extremes. The rising frequency of heatwaves has been associated with adverse health outcomes and increased energy demand in areas with significant population density [1], [2]. Environments characterized by limited vegetation, heat-absorbing materials, and extensive use of active air conditioning systems often experience elevated temperatures [3]. This dependence on mechanical climate control exacerbates local thermal conditions and poses challenges for maintaining resilient, sustainable, and equitable living spaces.

Addressing these challenges requires methods that capture the spatial and temporal variability of local microclimates. While high-fidelity computational simulations, such as multiphysics models based on CFD (Computational Fluid Dynamics), provide detailed insights, their high computational cost limits their practical application across multiple scenarios. Reduced Order Models (ROMs) have emerged as a promising solution

to enable faster approximations while maintaining sufficient physical accuracy.

Reduced Order Models (ROMs) are widely used to approximate high-dimensional physical systems with significantly reduced computational cost. ROMs are generally categorized into intrusive (IROM) and non-intrusive (NIROM) approaches [4], [5]. Intrusive ROMs work by manipulating the underlying physical equations of the system, such as the Navier-Stokes equations, to create a simplified version that is faster to solve. Techniques like Proper Orthogonal Decomposition (POD) [6] are often used to find a reduced space where the equations can be projected. Because intrusive methods start from the original governing equations, they can build reduced models that still reflect some of the system’s physical behavior. In contrast, Non-Intrusive ROMs rely mostly on data, without requiring access to or modification of the underlying simulation code. These models employ machine learning and statistical techniques to learn behavior of the physical system directly from simulation or experimental data. While NIROMs are generally easier to implement and can avoid potential numerical challenges of IROMs, they often need significant computational resources during training. Still, once trained, NIROMs can offer large speedups at prediction time, sometimes reducing simulation costs by several orders of magnitude compared to conventional CFD methods [7]. However, their reliance on data alone, mostly without explicit embedding of physical laws, may limit their robustness in certain engineering applications. The application of non-intrusive ROMs to urban microclimate problems has recently gained attention, with several studies focusing on airflow and turbulence prediction. [8] developed a non-intrusive ROM (NIROM) for predicting transient three-dimensional urban airflow. Their method applies Proper Orthogonal Decomposition (POD) [6] to high-fidelity CFD snapshots and trains a Gaussian Process Regression (GPR) model [9] to forecast the evolution of POD coefficients. Their NIROM reduced computational costs by six orders of magnitude but focused on forecasting small time-step changes in airflow under fixed boundary conditions.

[10] proposed a reduced order modeling approach for urban boundary layer flows by combining Singular Value Decomposition (SVD) with Recurrent Neural Networks (RNNs). High-fidelity training data were generated using filtered Navier–Stokes simulations, and an optimized Long Short-Term

Memory (LSTM) model was trained to predict the temporal evolution of the reduced flow structures. Their method accurately reconstructed turbulent kinetic energy (TKE) dynamics and velocity fields, achieving a computational speedup of approximately forty times compared to CFD simulations.

Most studies applying reduced order modeling methods in urban applications have focused on predicting immediate future states under very similar boundary conditions, and primarily on airflow alone. While effective for accelerating high-fidelity simulations, this narrow focus limits their practical deployment in real-world urban planning and design contexts. For meaningful assessments of microclimates for thermal comfort and energy resilience, it is equally important to model radiative heat transfer, which significantly influences human thermal exposure. Beyond reduced order modeling, machine learning approaches have also been explored for predicting thermal comfort. However, many of these studies focus on estimating overall comfort ratings or index values without resolving the spatial distributions of key microclimatic variables such as air temperature, wind speed, and mean radiant temperature [11]. A recent review further highlights that although machine learning has substantially improved predictive performance in thermal comfort studies, challenges remain in capturing spatial variability and fully accounting for radiative influences [12].

To address these gaps, the present study develops and evaluates a machine learning-based framework for predicting microclimatic variables pertinent to human thermal comfort within an idealized domain consisting of six modeled buildings. The variables of interest include air temperature, wind velocity, and mean radiant temperature, predicted across a wide range of external weather conditions.

The underlying physics problem involves modeling coupled airflow, heat transfer, and radiative processes in inhabited environments, typically governed by the Navier–Stokes and Energy Equations. While CFD simulations can resolve these phenomena accurately, they are too computationally intensive for practical multiseasonal studies. Machine learning offers a way to build ROMs that predict key variables pertinent to human thermal exposure efficiently based on patterns learned from high-fidelity simulation outputs.

II. METHODOLOGY

To develop a Reduced Order Model (ROM) capable of efficiently predicting urban microclimate variables across a wide range of weather conditions, a multi-stage framework is proposed. The methodology consists of three main stages: (1) generation of training data through high-resolution multiphysics simulations; (2) data preprocessing including dimensionality reduction and distribution rebalancing; and (3) machine learning model training to predict spatial fields of air temperature, wind velocity, and mean radiant temperature (MRT). The following sections describe the simulation setup, data processing pipeline, and machine learning models in detail, including how physical principles are embedded in the workflow.

A first step is to obtain Training Data for which the ROM should be optimized. The next section is introducing the multiphysics Simulations, which introduces the data source for the ROM.

A. Data Sources

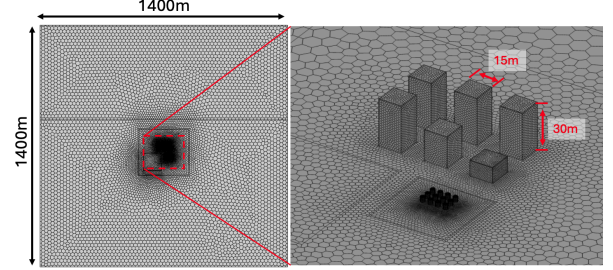


Fig. 1: Domain and mesh of the artificial urban environment investigated in this study.

1) *ROI and Time Period*: The ROM is developed using CFD simulations of an idealized urban environment consisting of six simplified buildings, tree zones, a water body, grass surfaces, and asphalt-covered roads. This synthetic geometry was selected to enable fast simulation turnaround during model development and prototyping. Figure 1 shows the domain and its mesh configuration. Output variables were extracted from a horizontal slice located 1.5 meters above the ground, corresponding to typical pedestrian height for thermal comfort analysis.



Fig. 2: Left: Location of the UPC campus within Barcelona. Right: Proximity of the UPC campus to the reference weather station (Fabra Observatory), used for deriving boundary conditions.

Although the domain geometry is artificial, the boundary conditions driving the simulations are based on real historical weather data collected near the Universitat Politècnica de Catalunya (UPC) campus in Barcelona, Spain. The ROM is trained to predict microclimatic fields for the period between March 1st and August 1st, 2024, matching the timeframe of a parallel sensor deployment carried out on the actual UPC campus. Figure 2 shows the location of the UPC campus and the selected weather station [13], which is situated approximately 1 km southwest of the campus on relatively neutral terrain. This proximity ensures that weather measurements

reflect boundary-layer conditions relevant to the simulation domain, supporting the long-term goal of transferring the ROM pipeline developed in this study to a real urban setting with dense sensor coverage for validation.

The weather station records wind speed, wind direction, air temperature, and global horizontal solar irradiance at a 30-minute interval. These measurements were linearly interpolated to an hourly resolution to match the timestep size of the CFD simulations. To identify representative days for the climate conditions across the chosen timeframes, a clustering analysis was conducted using K-Means. For characterizing daily weather patterns, the full diurnal profiles of each variable were organized into feature-vectors representing the temporal evolution over each day. K-means clustering was then applied to these daily profiles to group days with similar weather dynamics into six clusters. A representative day for each cluster was selected by identifying the daily profile closest to the corresponding cluster centroid. To complement these six typical cases, six additional days were selected to capture edge conditions within the study period. These edge cases were identified by selecting days with the highest and lowest daily averages of air temperature, wind speed, and solar irradiance. This ensured that rare combinations of input conditions likely to have a strong influence on the simulated microclimate were included in the training dataset, thus enhancing the ability of the ROM to generalize in a wider range of scenarios. An overview of the selected days relative to the daily profiles can be seen in Figure 3. "Cluster Days" indicate the representative days with normal conditions obtained through clustering. "Lowest-" and "Highest Extremes" indicate the days with lowest and highest daily average values for air temperature, wind speed and solar irradiation.

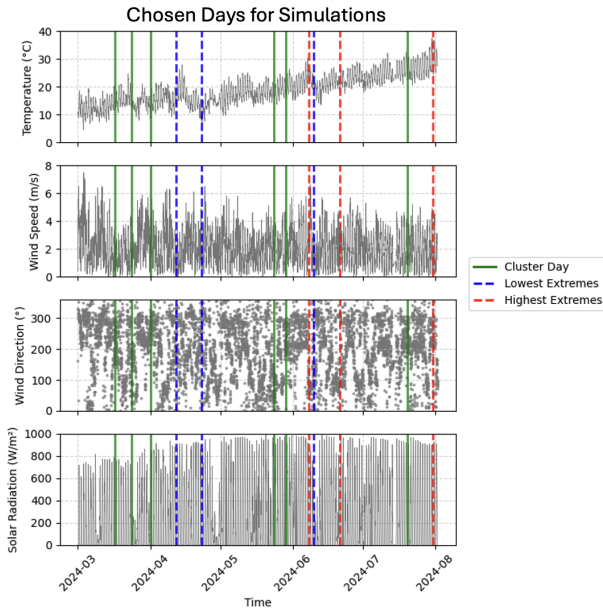


Fig. 3: Selected Days for simulation representing weather conditions across the chosen modelling period from March 1st to August 1st, 2024

For the 12 chosen days, CFD-based multiphysics simulations were carried out in the commercial code ANSYS Fluent 2024R2. The full physics phenomena occurring in outdoor micro climates are modelled, including radiative, conductive and convective heat transfer, including cooling from evaporation and shading effects from vegetation. The Reynolds-Averaged Navier-Stokes equations (RANS) were applied, with closure provided by the realizable $k - \epsilon$ turbulence model. The mesh has a resolution of 2.1 million elements, with element sizes ranging from 10 centimeters to 10 meters. The calculation was carried out on the NYU Greene High Performance Computing cluster and took approximately 6 hours to complete for a full day of hourly results. A single timestep of 1 hour took approx. 25 minutes to complete. The applied materials and surface properties, including thermal conductivity, heat capacitance, emissivity, and solar absorptivity, were assigned based on typical values for urban materials such as concrete, aluminum, glass, asphalt, vegetation, and water surfaces.

B. Data Preprocessing

Figure 4 presents an example of the postprocessed simulation output used as input for model training. Each training example consists of three grayscale images representing spatial fields of air temperature, wind speed (magnitude of velocity), and mean radiant temperature (MRT), extracted at 1.5 meters above ground level. Pixel intensities correspond to the local values of each variable. In addition to these spatial fields, five scalar input parameters are associated with each timestep: hour of the day, wind speed, wind direction, solar irradiance, and air temperature. The example shown corresponds to the following input values:

- Hour: 21.0
- Wind speed (m/s): 2.39
- Wind direction (°): 88
- Solar irradiance (W/m²): 3.0
- Air temperature (K): 290.35

In total, 288 such examples were generated, corresponding to 12 simulated days with 24 hourly outputs each. In the ROM workflow, the five scalar boundary-condition parameters (hour, wind speed, wind direction, solar irradiance, and air temperature) constitute the model inputs X . The model outputs Y are the three spatial fields produced by the CFD solver, mean radiant temperature (MRT), air temperature (AT), and wind velocity magnitude (MagVel), each sampled on the 1.5 m horizontal slice and stored as grayscale images.

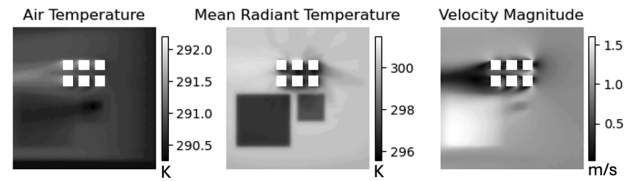


Fig. 4: Postprocessed example output from CFD simulations showing spatial fields of air temperature, wind speed, and MRT, used for ROM training.

The 5 scalar values (hour, wind speed, wind direction, solar irradiance, and air temperature) serving as input to the computation of a single timestep, show only low to moderate linear relationships, indicating that each parameter contributes distinct information for generating the output fields of mean radiant temperature (MRT), wind velocity magnitude (MagVel), and air temperature (AT). The strongest correlation, measured by the Pearson correlation coefficient (r), occurs between solar irradiance and wind speed ($r = 0.58$), suggesting that increased wind speeds tend to accompany periods of higher solar radiation. Additionally, a moderate correlation is observed between solar irradiance and air temperature ($r = 0.47$), consistent with expected physical interactions. Other feature pairs show only minor correlations, further highlighting the suitability of these parameters as independent inputs to the machine learning model.

The distributions of the five scalar input features across the entire dataset are illustrated in Figure 5. A clear imbalance is noticeable, particularly for solar irradiance and wind speed. Solar irradiance is heavily skewed toward zero because most simulated timesteps correspond to nighttime conditions, while wind speed is concentrated around moderate values between 2 and 3 m/s. Air temperature clusters near typical ambient conditions (287–293 K). Such imbalances can bias the machine-learning model toward commonly occurring conditions, reducing prediction accuracy under rarer permutations of the input parameters.

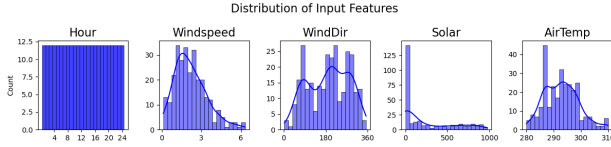


Fig. 5: Distribution of scalar input features across the entire dataset, highlighting imbalances particularly in solar irradiance and wind speed.

The identified imbalance in input feature distributions (Figure 5) can introduce bias into the predictive accuracy of the ROM, favoring conditions that appear frequently. To mitigate this bias and enhance the ROM’s ability to generalize across diverse conditions, a rebalancing strategy for the input training data was implemented, as described in the following section.

1) *Rebalancing of input data:* To ensure that the model generalizes well across a wide range of boundary conditions, a random 85%–15% split was applied to separate the dataset into training and testing subsets. This randomization prevents the test set from being biased toward particular types of weather conditions. The following rebalancing procedure was applied exclusively to the training subset, preserving the test set for unbiased evaluation and reliable performance assessment.

Imbalances in the scalar input features (Figure 5) revealed an over-representation of nighttime (zero-irradiance) and moderate-wind conditions. To reduce predictive bias toward these frequent weather states, the training data was

rebalanced by selectively duplicating samples representing rare input combinations.

Specifically, training samples were binned along three key variables: wind speed, solar irradiance, and wind direction parameters with the greatest influence on outdoor microclimate dynamics. Each feature was divided into five equal-width bins, and samples were grouped based on their joint bin membership, yielding up to $5^3 = 125$ possible bin combinations. Empty bins were ignored.

Next, the 80th percentile of sample counts across non-empty bins was computed to define a minimum bin size threshold, $n_{\min} = 7$. Bins containing fewer than $n_{\min}$ samples were upsampled via random duplication until they reached the threshold. Bins already meeting or exceeding the threshold were left unchanged. This procedure increased the number of training samples from 244 to 449, enhancing coverage of underrepresented weather scenarios.

Figure 6 shows the distribution of input features before (top row) and after (bottom row) rebalancing. Notably, the solar irradiance distribution becomes slightly more uniform, while the occurrence of higher wind speeds and elevated air temperatures increases, reducing skew and improving coverage of extreme conditions.

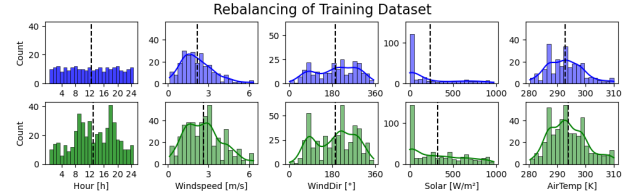


Fig. 6: Input-feature distributions in the training set before (blue) and after (green) rebalancing.

2) *Dimensionality Reduction via PCA:* Each of the three CFD-derived output fields (mean radiant temperature (MRT), air temperature (AT), and wind velocity magnitude (MagVel)) was exported as a 1000×1000 pixel grayscale image per timestep, resulting in 1 million dimensional output vectors. To reduce model complexity and training time, Principal Component Analysis (PCA) was applied to the balanced training set.

Figure 7 shows the cumulative explained variance for each of the three output fields under the balanced dataset. MRT and AT required relatively few components to surpass the 99.99% threshold, while MagVel exhibited more spatial variability, requiring the full 25 components explaining 99.98%.

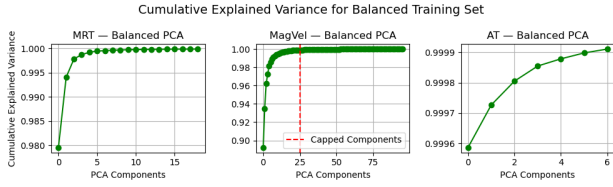


Fig. 7: Cumulative explained variance by PCA components for each output field (MRT, MagVel, AT) using the balanced dataset. A cutoff at 25 components was enforced for computational efficiency.

The resulting PCA-transformed outputs form the reduced-dimensional target vectors for model training. These transformations were also applied to the test set using the same PCA models to ensure consistent dimensionality during evaluation.

While PCA retained at least 99.99% of the variance for each field, this compression still introduces a slight trade-off in spatial fidelity. The retained components capture dominant modes of variability but may under-represent localized features that contribute little to total variance. As a result, reconstructed fields may smooth out fine-scale patterns.

C. Machine Learning Models

After PCA-based dimensionality reduction, three separate regressors were trained to model the reduced representations of mean radiant temperature (MRT), air temperature (AT), and wind velocity magnitude (MagVel). Each model receives the same 5-dimensional scalar input vector (hour, wind speed, wind direction, solar irradiance, and air temperature) and predicts a low-dimensional output vector corresponding to the respective PCA-transformed field.

1) *Random Forest Regression*: Random Forest models were used to predict the PCA-compressed representations of air temperature (AT) and wind velocity magnitude (MagVel). These models were chosen due to their ability to capture complex nonlinear relationships and their robustness against overfitting with relatively few samples. A Random Forest is an ensemble method that averages predictions from multiple decision trees trained on bootstrapped subsets of the data, which helps reduce variance and improve generalization. Unlike linear regression, which failed to capture spatial variability in preliminary testing, Random Forests provided strong baseline performance without requiring extensive feature engineering.

Hyperparameters were optimized using five-fold cross-validation with a grid search across tree depth, number of estimators, and minimum split size. The optimization objective was to minimize the mean absolute error (MAE), which provides a physically interpretable metric by directly quantifying the average deviation in real world units. MAE was favored over the coefficient of determination (R^2), which, while useful for assessing explained variance, can be misleading in microclimate prediction, particularly when target variables exhibit low variability or when predictions capture general trends but not accurate magnitudes. Using MAE supports the connection between model performance and physical realism in the predicted fields.

To further mitigate overfitting, especially in the air temperature model where one input feature was suspected to dominate, Gaussian noise (standard deviation ± 0.5 K) was added to the input air temperature during training. This regularization strategy was introduced to encourage broader feature utilization and improve generalization.

2) *Gradient Boosting for MRT*: While the Random Forest model achieved strong performance for predicting the fields for air temperature and wind velocity magnitude, it was less effective for mean radiant temperature (MRT), which is influenced by more complex radiative interactions. This motivated the use of a different regression approach for MRT prediction.

A Gradient Boosting Regressor was employed, as it is similarly capable of modeling nonlinear relationships but differs in its learning strategy. Gradient Boosting builds an ensemble of shallow decision trees sequentially, with each tree correcting the residual errors of its predecessors. This iterative process allows the model to gradually refine its predictions, making it better suited for capturing the nuanced spatial patterns present in MRT fields.

Hyperparameters were optimized using five-fold cross-validation with grid search across the number of trees, learning rate, and maximum tree depth. The objective remained the minimization of mean absolute error (MAE), chosen for its direct physical interpretability.

III. RESULTS

1) *Final Model Training and Evaluation*: The final ROM was constructed by retraining each of the three best-performing models, using the full balanced training set and the optimal hyperparameters identified during cross-validation. A Gradient Boosting Regressor was used for MRT, while Random Forest Regressors were used for MagVel and AT. Table I summarizes the selected hyperparameters, and Table II reports performance on the 15% held-out test set.

TABLE I: Best hyperparameters used for the final ROM components

Output Field	Model Type	Selected Hyperparameters
MRT	Gradient Boosting	Number of Trees = 200 Maximum Tree Depth = 5 Learning Rate = 0.1
MagVel	Random Forest	Number of Trees = 300 Maximum Tree Depth = 30 Min Samples per Split = 2
AT	Random Forest	Number of Trees = 200 Maximum Tree Depth = 10 Min Samples per Split = 4

TABLE II: Test performance metrics of final models

Output Field	MAE	R^2
MRT	1.219 K	0.875
MagVel	0.137 m/s	0.961
AT	0.381 K	0.994

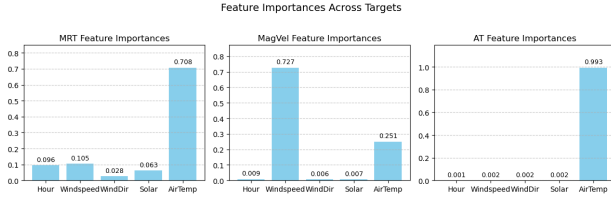


Fig. 8: Feature importance after training of the models.

The Air Temperature (AT) model achieved the highest overall performance, with an R^2 of 0.994 and a MAE of 0.381 K. These values reflect the final model after regularization. To mitigate overfitting—particularly the model’s near-exclusive reliance on the air temperature input (99.3%) as shown in Figure 8—Gaussian noise with a standard deviation of ± 0.5 K was added to the air temperature input during training. While this noise injection slightly reduced test performance compared to the unregularized model, it is assumed to improve generalization by marginally redistributing feature importance and reducing reliance on a single dominant predictor.

The wind velocity magnitude (MagVel) model also performed strongly ($R^2 = 0.961$, MAE = 0.137 m/s), with a more balanced feature importance. It relied primarily on wind speed (72.7%) and air temperature (25.1%), consistent with the moderate positive correlation ($r = 0.38$) observed between these two variables in the training data. Surprisingly, wind direction had very low influence (0.6%) despite its clear impact on simulated flow patterns. This may indicate that the model captures directional effects indirectly, but does not rely heavily on wind direction as an explicit input due to limitations in how directional features are handled.

The MRT model, which involved more complex radiative interactions, showed the lowest—but still strong—performance ($R^2 = 0.875$, MAE = 1.219 K). Feature importance was more evenly distributed, with contributions from air temperature (70.8%), windspeed (10.5%), and hour (9.6%). Compared to a prediction of MRT via Random Forest, the Gradient Boosting model yielded modest but consistent improvements in both R^2 and MAE for this variable, demonstrating its value for learning subtle .

The mean radiant temperature (MRT) model showed the lowest, but still strong performance among the three targets, with an R^2 of 0.875 and a MAE of 1.219 K. As seen in Figure 8, its feature importance was more evenly distributed than for the other models, with notable contributions from air temperature (70.8%), wind speed (10.5%), and hour of day (9.6%). The use of Gradient Boosting for this variable yielded modest but consistent improvements over a Random Forest baseline, particularly in terms of reduced MAE.

The mean radiant temperature (MRT) model exhibited the lowest, but still robust performance among the three outputs, with an R^2 of 0.875 and a MAE of 1.219 K. As shown in Figure 8, its feature importance was more evenly distributed than for the other models, with substantial contributions from air temperature (70.8%), wind speed (10.5%), and hour of day (9.6%). Compared to an earlier Random Forest model

for MRT prediction ($R^2 = 0.867$, MAE = 1.281 K), the Gradient Boosting approach delivered modest but consistent improvements, especially in terms of MAE. These subtle yet significant gains highlight the implemented Gradient boosting model’s advantage in capturing the more diffuse and nonlinear radiative patterns that shape MRT, where no single input fully governs the output variability.

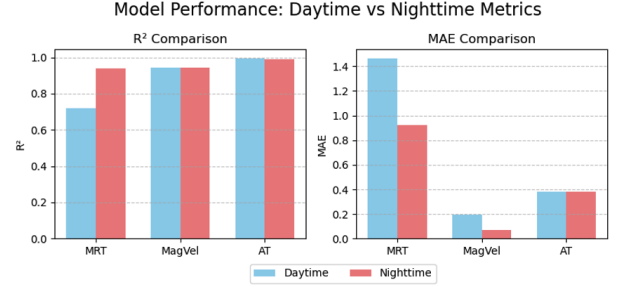


Fig. 9: Model performance comparison across daytime and nighttime samples. MRT shows higher error and lower R^2 during daytime, while AT and MagVel remain relatively stable.

2) *Performance by Time of Day*: Model performance varied notably between daytime and nighttime, especially for mean radiant temperature (MRT). Figure 9 compares R^2 and MAE across both times. MRT exhibited a clear drop in R^2 and increase in MAE during the day, reflecting the added complexity of radiative interactions under solar exposure. During daytime hours, solar loading, reflective interactions from different surface materials, and building orientations introduce localized effects that are harder to generalize from scalar inputs alone. At night, the absence of direct solar radiation reduces this variability, leading to improved model agreement with ground truth.

The wind velocity magnitude (MagVel) model maintained high R^2 across both regimes but exhibited a slightly higher MAE during the day. This may be due to more irregular and stronger wind patterns associated with buoyancy-driven turbulence, which are more prevalent during daylight hours.

Air temperature (AT) prediction was stable between day and night, with nearly identical MAE values and consistently high R^2 . This is partly attributable to the relatively low spatial variance of air temperature in each individual simulated timestep and the limited range of input air temperatures across the dataset. Moreover, the high feature importance of the air temperature input (99.3%) explains much of the field-level output, leading to a high ratio of explained variance.

3) *Example Predictions*: To illustrate how the model performs under specific boundary conditions, Figure 10 shows an example prediction for a nighttime test sample (Input values: Hour = 24, Wind Speed = 1.69 m/s, Wind Direction = 84° (from East), Solar Irradiance = 0 W/m², Air Temperature = 290.05 K). Ground truth and predicted fields for MRT, velocity magnitude, and air temperature are shown alongside their respective spatial error maps.

In the shown example, the predicted MRT field slightly underestimates the overall radiation levels, resulting in a domain-wide offset of approximately 1 K. This discrepancy is expected at late evening hours, when MRT typically remains elevated above air temperature due to heat trapped in the surface materials which accumulated during the day. The spatial structure of the MRT field is well preserved: the water body (large square in the lower left) and the grass-covered zone with tree canopies (smaller square to the right) are both correctly predicted as cooler regions.

The velocity field prediction captures the broader flow structure well, including the wake zones downstream of buildings and the elevated wind speeds above the water body, where surface roughness is minimal. Despite the model assigning low feature importance to wind direction, directional flow patterns are clearly maintained. Discrepancies in the velocity error map are limited to small, spatially scattered deviations near obstacles and in shear zones, without consistent over- or underprediction trends.

Air temperature predictions closely match the ground truth. The model successfully reproduces spatial features such as higher temperatures in the wake of buildings and cooler regions downstream of trees, driven by evaporative cooling. While the largest differences appear around the tree zones, they remain below ± 0.1 K, indicating that the model effectively captures both radiative and convective influences on local air temperature.

Comparison Ground Truth vs Prediction

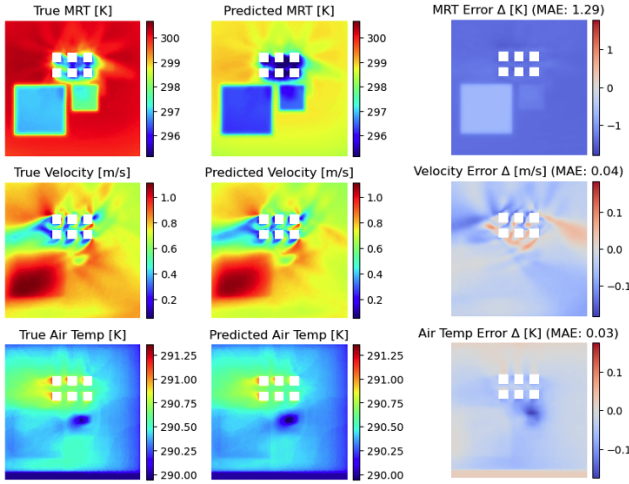


Fig. 10: Comparison of ground truth and predicted fields for a nighttime test sample (Hour = 24, Wind Speed = 1.69 m/s, Wind Direction = 84°, Solar Irradiance = 0 W/m², Air Temperature = 290.05 K). Columns show: ground truth, model prediction, and spatial error for MRT, wind velocity magnitude, and air temperature.

4) *Error Trends Across Input Features:* To explore how model accuracy varies under different conditions, the mean absolute error (MAE) was plotted against each of the five scalar inputs for all test samples. Figure 11 shows these

relationships, with trend lines in red highlighting the general error behavior across the input ranges.

MRT predictions show only weak and inconsistent trends. Although the MAE tends to be slightly elevated around midday hours and marginally lower at higher air temperatures, these patterns lack a clear or consistent rule. This reflects the inherent complexity of MRT, which is governed by nonlinear and spatially diffuse radiative processes that cannot be easily inferred from scalar inputs alone. As such, the model struggles to consistently reduce error in these cases despite the overall strong performance.

By contrast, wind velocity magnitude (MagVel) exhibits clear and interpretable patterns. MAE increases strongly with wind speed, likely due to a lower representation of high-speed cases in the training set, as seen in the input feature distributions (Figure 6). The model struggles more with rare high-wind scenarios, particularly those exceeding 4.5 m/s. A second upward trend in MAE with rising air temperature may reflect the limited number of training samples above 300 K, again suggesting an underrepresentation of these edge cases during training.

Air temperature (AT) predictions remain relatively robust across input conditions, though a mild trend appears with respect to the air temperature input itself. Unexpectedly, errors are somewhat larger for lower air temperature values (e.g. $\downarrow$ 285 K), while predictions improve at higher values. One possible explanation is that high-temperature scenes tend to exhibit more spatial uniformity, making them easier to regress accurately. Additionally, there are more training samples above 300 K than in the extreme low-temperature range, which likely contributed to improved performance for this input data range.

IV. DISCUSSION

The results indicate that tree-based machine learning models, when trained on CFD-derived simulation data, can predict high-resolution microclimate fields such as air temperature, wind velocity magnitude, and mean radiant temperature (MRT) with significant accuracy. Random Forests captured well-structured outputs like air temperature and wind speed, while Gradient Boosting showed advantages in modeling the more diffuse and nonlinear spatial patterns of MRT. These findings support the viability of data-driven Reduced Order Models (ROMs) as fast and detailed surrogate models for evaluating human thermal exposure across diverse environmental conditions. Importantly, the trained ROM generates predictions for all three fields in approximately 7 seconds, compared to roughly 25 minutes per timestep using the original CFD solver, yielding a speedup of over 200 times without substantial loss in physical fidelity.

Despite the strong performance metrics, several challenges and limitations remain. The exceptionally high R^2 values observed for air temperature (AT) and wind velocity magnitude (MagVel)—particularly under low-variance conditions—suggest that this metric may overstate the model's practical accuracy. This reinforces the use of mean absolute error (MAE) as a more physically grounded performance

Mean Absolute Error per Input Feature

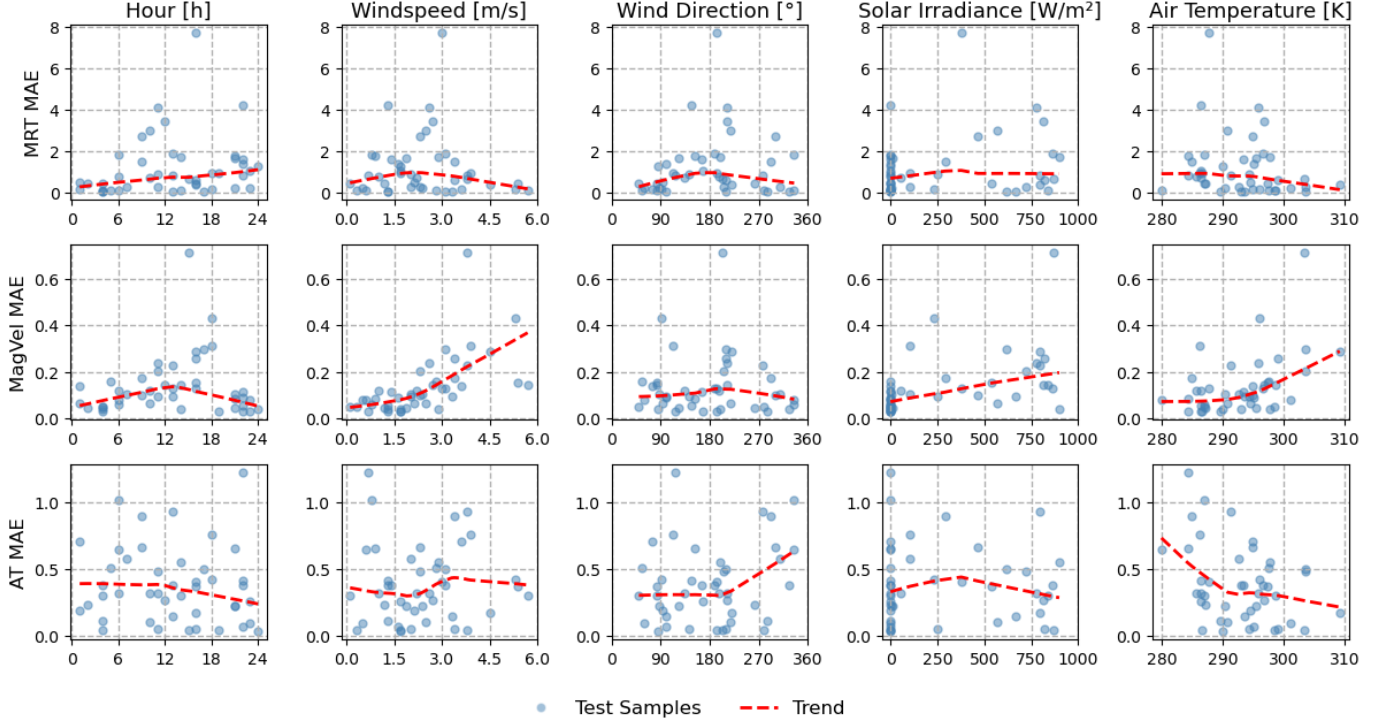


Fig. 11: Mean absolute error (MAE) per test sample plotted against each scalar input variable. Red dashed lines indicate trends.

measure, as it directly quantifies the average prediction error in the same units as the target variable and is less distorted by low target variability or narrow distributions.

Error trend analysis further revealed that model performance depends strongly on the density and distribution of training examples across input conditions. For instance, MRT predictions were less accurate during the day, when radiative interactions become more complex due to direct solar loading, reflection, and surface material properties. Similarly, the MagVel model struggled more with higher wind speeds, which were under-represented in the training dataset despite rebalancing efforts. These findings underscore the importance of incorporating more edge cases and extreme conditions into the training data to improve generalization.

In addition, the feature importance analysis revealed a structural limitation in how wind direction was represented. Although wind direction clearly influences flow behavior, it contributed minimally to the model’s predictions, likely due to its representation as a single scalar value without directional context. To address this, spatial masks highlighting wakezones based on building geometry and prevailing wind directions, were tested as additional input features. However, their inclusion did not yield measurable improvements in prediction accuracy. This outcome suggests that more advanced learning architectures, such as convolutional neural networks, may be better equipped to capture directional and spatial dependencies from raw input patterns than the tree-based model structure

used in this study.

Lastly, although the models are trained on physically grounded outputs from CFD simulations, they do not explicitly enforce physical laws during inference. Because tree-based models lack a loss-function framework suitable for embedding physical constraints, future efforts could explore deep learning approaches where conservation laws (for mass or energy) are integrated directly into the loss function. Such physics-informed architectures would offer a pathway toward more physically reliable predictions, especially under extreme or extrapolated input conditions.

V. CONCLUSION

This study developed a Reduced Order Model (ROM) based on Random Forest and Gradient Boosting regressors to predict spatially resolved urban microclimate variables using high-fidelity CFD simulation data. The key findings and proposed future directions are summarized below:

- **Good predictive performance:** The ROM achieved strong performance across all three target fields: air temperature (AT), wind velocity magnitude (MagVel), and mean radiant temperature (MRT), with R^2 values of 0.994, 0.961, and 0.875 respectively, and corresponding low MAEs.
- **Significant speedup:** Inference time was reduced from 25 minutes per CFD timestep to just 7 seconds using the ROM, enabling efficient multi-scenario testing for planning applications.

- **Interpretability via MAE:** Mean Absolute Error (MAE) served as a more robust evaluation metric than R^2 , particularly in low-variance conditions where R^2 can exaggerate accuracy.
- **Error sensitivity to training data distribution:** Model performance deteriorated under high solar loads and strong winds, which are conditions underrepresented in the dataset, highlighting the need for more edge-case training data.
- **Underutilized wind direction information:** Wind direction, given as a single scalar, had little predictive impact on the wind speed field. Attempts to introduce wake zone masks showed no performance gains. More expressive representations and deep learning models (e.g. CNNs) may better capture directional dependencies.
- **Lack of explicit physical enforcement:** Although the ROM is trained on CFD data that reflect full physical processes, the models themselves do not impose physical laws during inference. Future work should explore neural network architectures with physics-informed loss functions that enforce constraints such as energy or mass conservation.
- **Future Work:** Future improvements include expanding the training dataset with additional edge conditions, testing autoencoders for unsupervised spatial feature extraction, evaluating the benefit of skipping PCA or standardizing inputs, and ultimately applying the ROM workflow to a real-world domain (UPC campus Barcelona) where a more comprehensive validation of predictions with sensor data is possible.

These findings suggest that the presented ROM framework offers a practical and scalable tool for predicting human-relevant thermal conditions at great resolution, providing a foundation for near real-time microclimate analysis in support of climate-resilient urban design.

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